



AI-Driven Innovations to Enhance Tea Production and Quality in Sri Lanka's Mid-Country

Project ID: 25-26J-133

OUR TEAM



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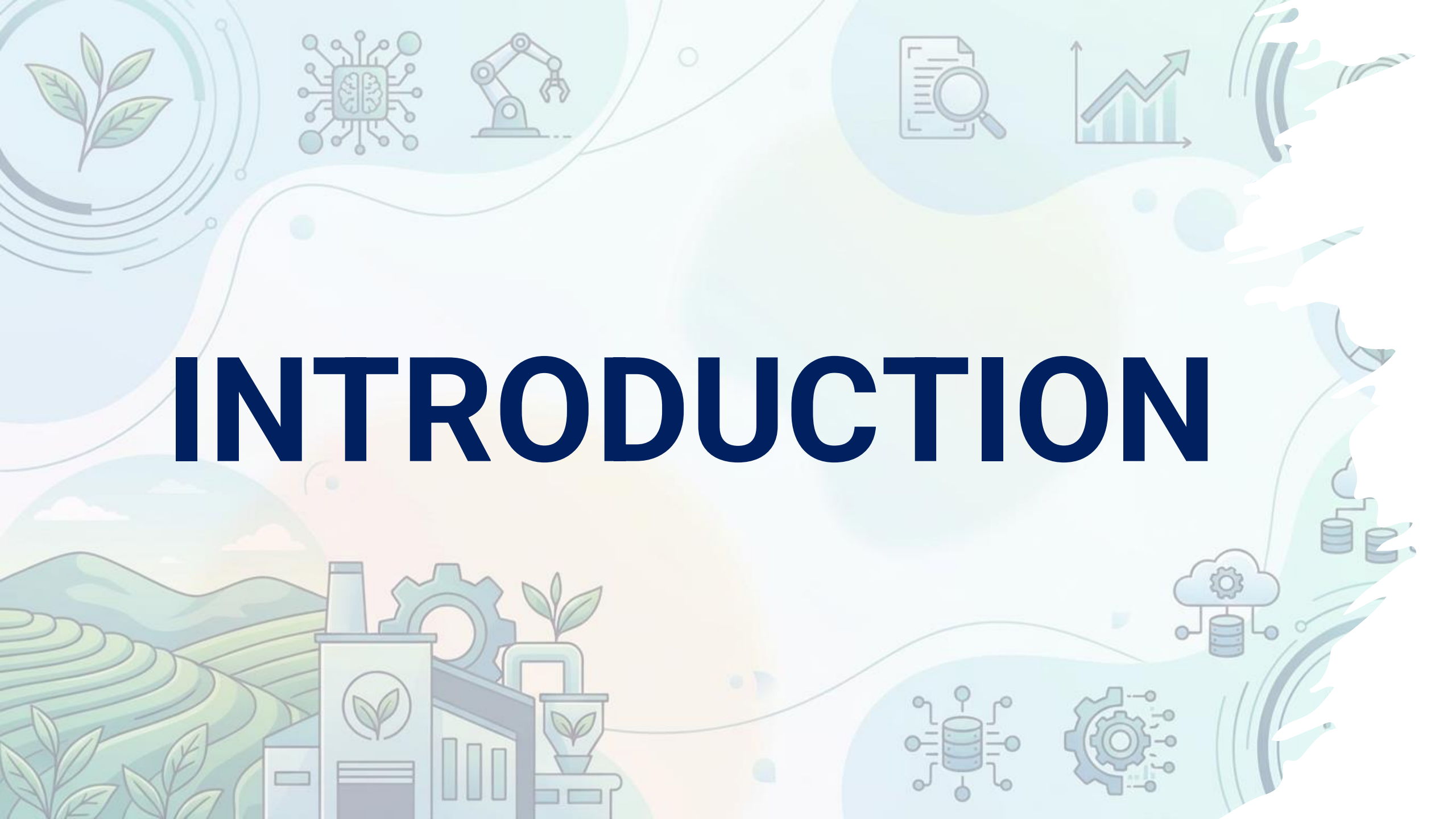
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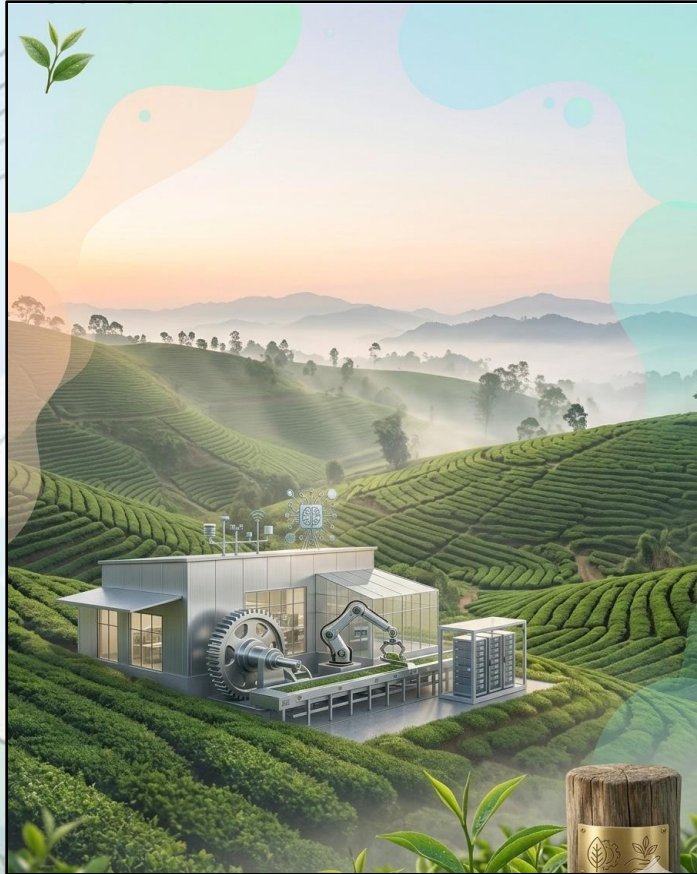
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INTRODUCTION

INTRODUCTION

Tea Vision– AI Driven Innovations to Enhance Tea Production and Quality in Sri Lanka's Mid-Country



- Sri Lanka's tea industry relies heavily on slow, paper-based processes that reduce efficiency.
- Manual methods cause inconsistent grading, poor yield tracking, unpredictable auctions, and late disease detection.
- We propose an AI system to automate the entire tea lifecycle, boosting productivity from the farm to the market.



Problem

PROBLEM STATEMENT & STAKEHOLDERS

The Sri Lankan tea industry, though vital to the economy, struggles with productivity, quality, and sustainability. Farmers face inconsistent plucking, quality loss, and undetected disease and pest outbreaks, while factories and auction stakeholders depend on manual grading, limited forecasting, and inefficient planning, leading to reduced competitiveness.

Stakeholders:

Primary users - tea farmers, estate managers, factories.

Secondary users - brokers, buyers, researchers.

Decision makers - estate owners, factory managers, and industry stakeholders.



RESEARCH GAP

Research Gap & Objectives

Research gap: Existing tea-sector solutions are fragmented. Most focus only on one area such as grading, forecasting, or disease detection, and do not provide an integrated platform for tea quality assessment, yield analytics, auction planning, treatment guidance and seasonal disease prediction.

Main objective: To design and validate an AI-driven decision-support platform for the Sri Lankan tea industry that improves tea quality evaluation, digitizes field records, predicts yield and market behavior, and supports early disease detection with actionable recommendations.



RESEARCH NOVELTY

RESEARCH NOVELTY

The novelty of this research lies in the fact that no similar end-to-end AI system currently exists for the tea sector in Sri Lanka. Although isolated models have been developed for specific agricultural tasks, this study is unique as it seamlessly integrates four distinct factors across the tea production lifecycle.

Additionally, each individual component of the system introduces its own novelty and innovation.



PROPOSED OVERALL SOLUTIONS

Solution Overview

Solution summary: We developed an integrated AI-based platform to support tea quality grading, yield forecasting, market demand analysis, and tea leaf disease detection for Sri Lanka's mid-country tea sector.

The Proposed Solution is Divided into Four Key Components:

1. Automated freshness grading and tea type classification
2. Yield logbook digitization and monthly yield forecasting
3. Tea market demand forecasting and smart auction planning
4. Tea Leaf Diseases, Invasive Species Detection with Treatments and Seasonal Disease Risk Prediction

1. Automated Freshness Grading and Tea Type Classification

- **Data & Segmentation:** Custom *TeaVision* Android app utilizing a dual Roboflow + HSV pipeline for precise leaf isolation.
- **Freshness Grading:** Extracted 25 visual features; an **MLP Neural Network** achieved the top accuracy (**86.67%**) out of 10 tested models.
- **Damage Detection:** **Roboflow V4** combined with **SAHI** (slicing) successfully detected 7 TRI-defined classes and small damage spots.
- **Advisory Module:** Hybrid RAG pipeline (Sentence-BERT + BM25) across 156 TRI documents delivered near-perfect retrieval (Hit Rate/MRR: **1.0**).

2. Tea Yield Book Digitization and Predictive Analytics for Crop Yield Management

- **Data Acquisition & Alignment:** Interactive perspective rectification and manual boundary controls to eliminate geometric distortion, ensuring perfectly flattened, noise-free records for extraction.
- **Handwritten Yield Book Digitization:** Fine-tuned a cell-level PaddleOCR recognizer on labeled handwritten yield logbook cell images, transforming logbook photos into clean, structured Excel tables ready for analysis.
- **OCR Accuracy:** Achieved 65.0% exact cell-level accuracy for Tea yield book digitization module.
- **Monthly Yield Forecasting Model:** Achieved around 29% MAPE with SARIMAX, significantly improving prediction reliability over SARIMA and Prophet.

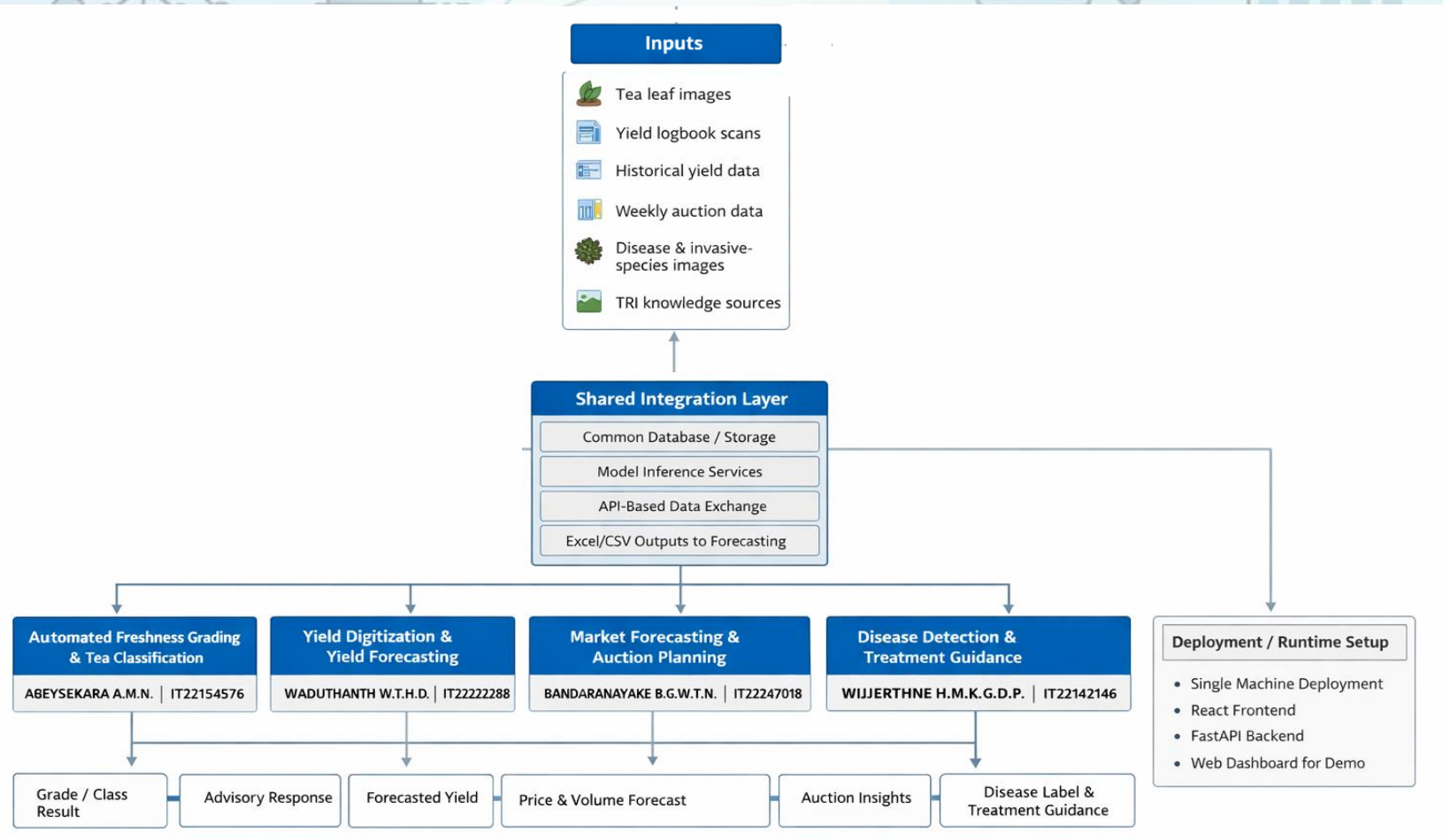
3. Tea Market Demand Forecasting and Smart Auction Planning

- **Massive Scale Forecasting:** Evaluated 1.5M Colombo Tea Auction records (2023–2025) using ARIMA, SARIMAX, and Prophet models.
- **Price Prediction Winner:** ARIMA achieved the highest accuracy, with MAPE error rates between just 4.28% and 5.11% across all elevations.
- **Smart Auction MAS:** Built an RL-based Multi-Agent System (Buyer, Factory, Broker) with SHAP and LIME for transparent AI decision-making.
- **Simulation Success:** Achieved a 96.0% auction clearance rate and 100% directional decision accuracy in dynamic market scenarios.

4. Tea Leaf Disease, Pest, Invasive Species Detection with Treatments and Seasonal Disease Risk Prediction

- **Disease & Pest CNN with Treatment Guidance:** Custom CNN trained on 6,000+ augmented images (6 classes). Achieved a highly robust test accuracy of 93.90% with Grad-CAM visualization.
- **Invasive Species CNN:** Second deep learning model successfully identified 5 common invasive weeds, achieving 91.49% test accuracy.
- **Seasonal Forecasting:** Random Forest and Gradient Boosting models analyzed 15,000 environmental records (temp, humidity, etc.), yielding 73% prediction accuracy.
- **Intelligent Assistance:** Integrated a NLP chatbot with Speech AI to provide real-time treatment guidance based on detected pathologies.

SYSTEM ARCHITECTURE & INTEGRATION



The platform integrates image-based AI, OCR-driven data digitization, forecasting models, retrieval/advisory methods, and user-facing interfaces into one decision-support ecosystem.



COMMERCIALIZATION & IMPACT

COMMERCIALIZATION & IMPACT

- **Ability of Commercialization:** High, as a modular Software-as-a-Service (SaaS) and mobile platform (TeaVision) designed for Regional Plantation Companies (RPCs), private tea estates, and brokering firms. It offers a clear path to industry-wide scaling across Sri Lanka following successful mid-country estate pilots.
- **Target Market & Value Proposition:** Primary customers are Regional Plantation Companies (RPCs), factory managers, and brokers. We empower them to maximize yields and auction profits by replacing manual, paper-based workflows with AI-driven crop diagnostics, automated logbook digitization, and multi-agent market forecasting.
- **Revenue & Cost Model:** Initial costs are recovered through a tiered B2B annual subscription. This fee is easily justified by the system's strong ROI: reduced disease-related crop loss, eliminated manual data-entry overhead, and optimized auction profit margins.

Demo Plan

Demo flow:

- Step 1 - Capture tea leaf image and show freshness grading / advisory output
- Step 2 - Upload disease image and show disease detection with treatment guidance / invasive species detection and seasonal prediction
- Step 3 - Scan a yield logbook page and show OCR-to-structured-data conversion And show monthly yield forecasting using digitized data
- Step 5 - Show tea market demand / auction prediction dashboard

Team handoff:

- Abeysekara A.M.N. – IT22154576- image capture, grading, advisory
- Waduthanthri W.T.H.D – IT22222268- OCR digitization and yield forecasting
- Bandaranayake B.G.W.T.N. – IT22247018- market demand forecasting and auction planning
- Wijerathne H.M.K.G.D.P. – IT22142146- Tea Leaf Disease, Invasive Species Detection with Treatments and Seasonal Disease Risk Prediction

CURRENT COMPLETION STATUS

Approximately 90% of the project has been completed. All major component's Functions have been developed and integrated to the overall system, and testing is currently pending

90%

DEMO PLAN

- ✓ Video Demonstration of the Fully Integrated System
- ✓ Each team member briefly explains their component and demonstrates it with a real-time demo.

  Best picking time is early morning when leaves have peak moisture.

 Quick Actions



Quick Leaf Scan

Take a photo and instantly identify leaf types, freshness, and damage



Grade My Tea

Upload a photo to get an AI-powered quality assessment



Disease Check

Scan a leaf to check for blister blight, red rust, or other diseases



Yield Forecast and Digitization

See predicted tea yield for your fields next month and yield book digitize



Market & Auction Strategy

Simulate auctions and forecast tea prices to maximize your profit



 Research Objectives

OBJ 1

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Tea Freshness Grading & Classification

Automated quality assessment using computer vision and ML

[Inspect Leaves →](#)

[Grade Quality →](#)

[Tea Handbook →](#)

OBJ 2

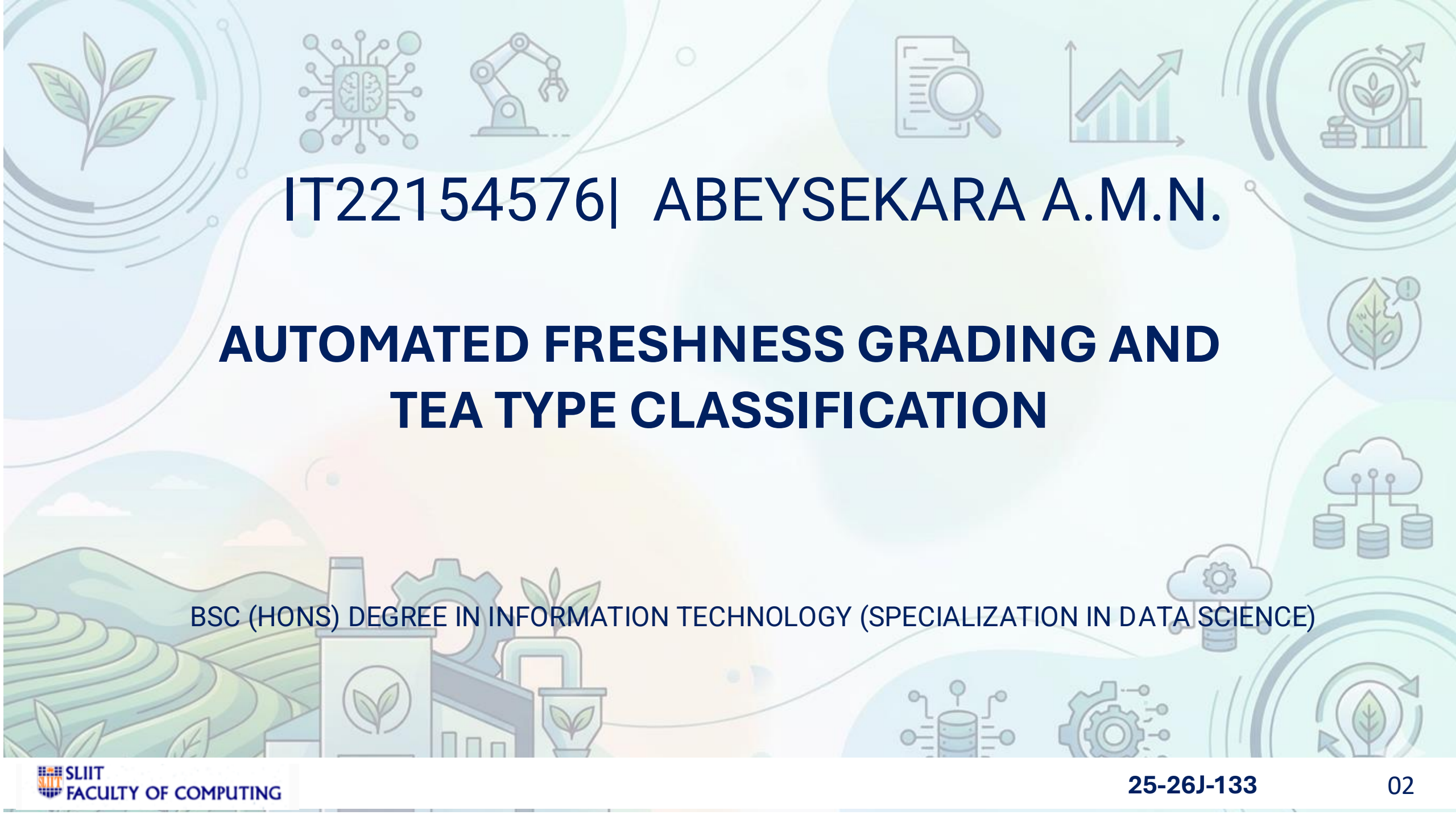
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Tea Yield Forecasting & Digitization

Logbook OCR and monthly yield forecasting

[Yield Forecast →](#)

[Digitize Logbook →](#)



IT22154576| ABEYSEKARA A.M.N.

AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY (SPECIALIZATION IN DATA SCIENCE)

AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION

Problem:

- Manual tea leaf inspection is time-consuming and subjective
- Inspector fatigue can lead to inconsistent grading decisions
- Accepting poor raw material lowers manufactured tea quality
- Small but important leaf details are difficult to assess reliably
- Existing systems often lack transparency and user trust
- Practical tea knowledge is not readily available in the field

Novelty : This study proposes a smart tea leaf analysis framework using Roboflow and YOLOv8 to identify important visual classes in fresh tea leaves. It also supports freshness grading through interpretable leaf features, includes image quality validation for reliable analysis, and provides a knowledge-based support component for better tea quality assessment and decision-making.

AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION

1. Data Collection (The Mobile App)

- **What it does:** This is the **TeaVision App** used by farmers in the field.
- **The Key Feature:** It uses the phone's camera and sensors to ensure the photo is taken at the perfect angle (less than 15° tilt) for accurate AI analysis.
- **The Action:** It sends high-quality images and metadata to the server.

2. Backend AI Server (The Brain)

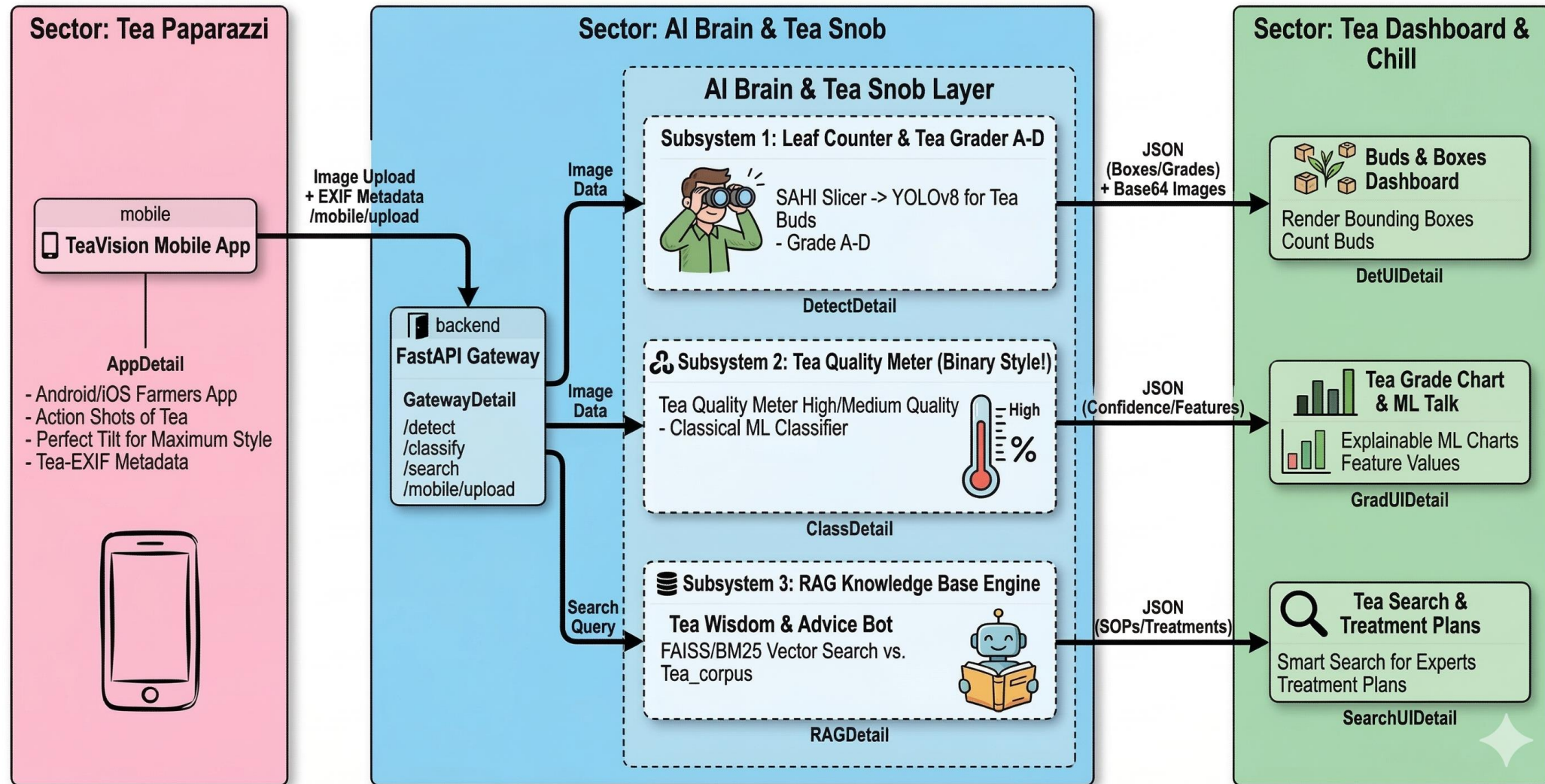
- **What it does:** This is where the heavy lifting happens using three specialized AI "engines":
 - **Detection Engine:** Uses **ROBOFLOW_V4** to find every single leaf and bud in the photo and count them.
 - **Quality Engine:** Analyzes colors and textures to grade the tea as High or Medium quality.
 - **Knowledge Engine:** A "Smart Search" (RAG) that looks through a digital library to find the best treatment plans for any issues found.

3. Frontend Dashboard (The Control Center)

- **What it does:** A web-based dashboard for managers and experts to view the results.
- **The Results:** It displays clear maps of the detected leaves, easy-to-read charts on quality grades, and expert advice for the farmers.

AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION

Tea Analysis System: Simple & Funny Edition



AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION

Completed after PP1:

- Re-labeled and improved the tea leaf image dataset
- Re-evaluated the detection approach due to weak YOLO performance
- Trained and tested an improved model for better detection accuracy
- Integrated the developed component into the TeaVision application
- Completed the end-to-end connection between model output and app workflow

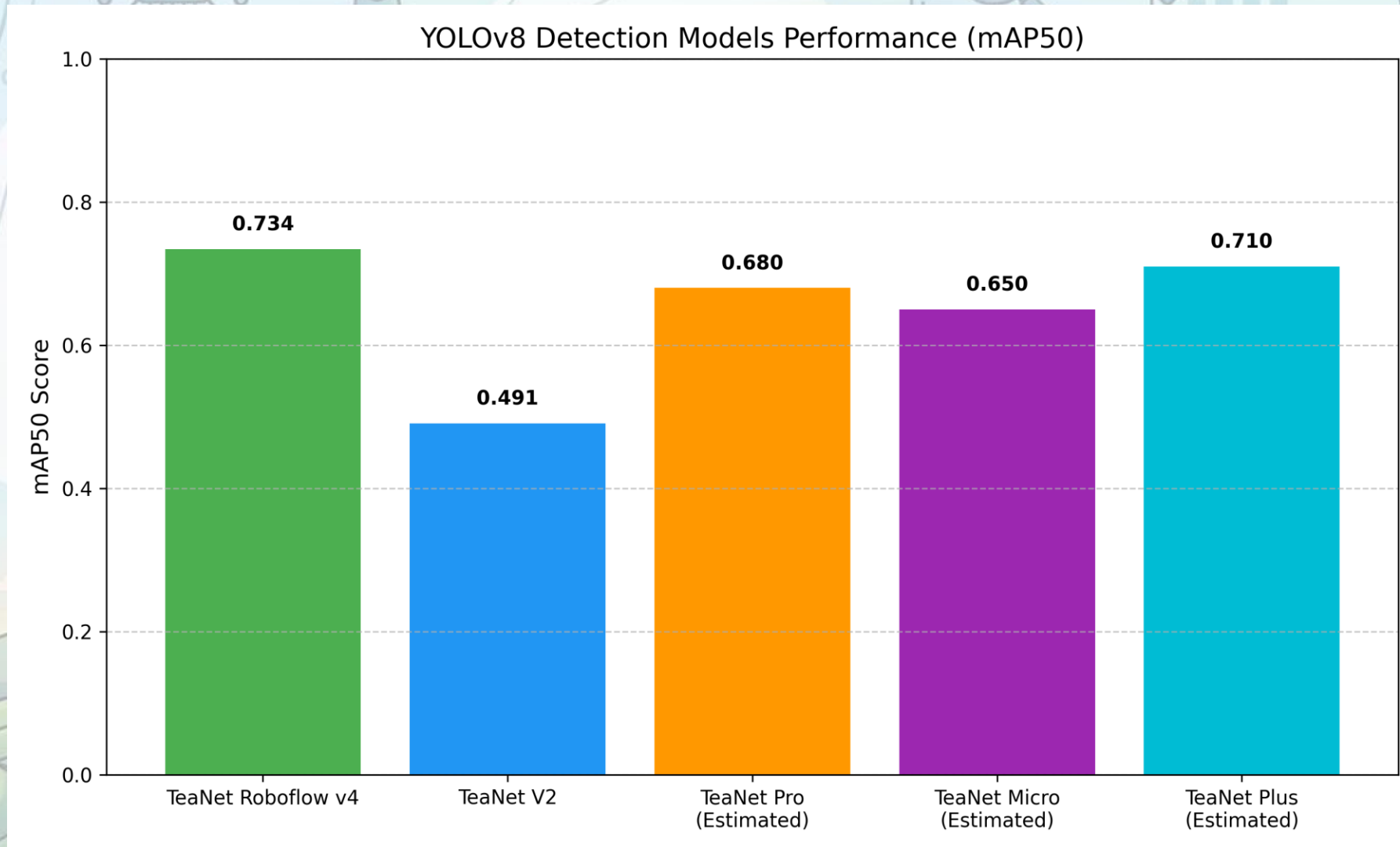
Changes done for PP1 recommendations

- Refined the dataset with additional and corrected annotations
- Moved beyond the initial YOLO-only approach to improve results
- Improved model performance through retraining and model adjustment
- Strengthened system integration for practical application use

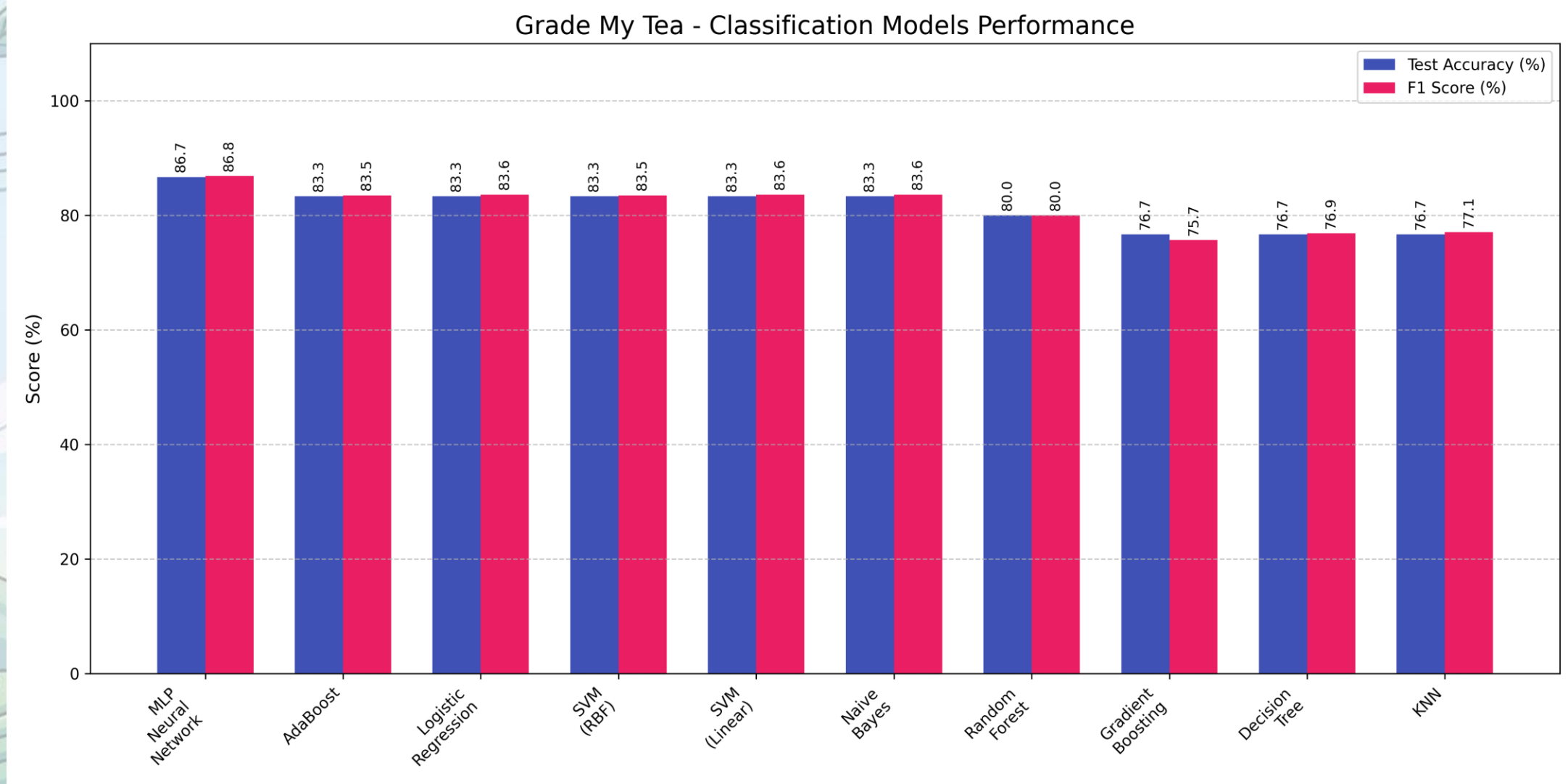
To Do

- We plan to improve model accuracy with more balanced data, optimize performance in real-world conditions, test with more tea leaf samples and environments, enhance the TeaVision interface and result explanations, and finalize the evaluation and presentation outputs.

AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION



AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION



AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION

Problems and Solutions

Undisclosed Grading (Black-Box AI): Farmers couldn't trust AI grading. **Solution:** Built an **Explainable ML System ("Grade My Tea")** that uses math (like color and shape indexes) to transparently prove exactly why a leaf got its grade.

Losing Tiny Details: Standard AI shrinks images, making it miss small tea buds. **Solution:** Added **SAHI to Roboflow**. It slices the image into a grid to detect microscopic buds accurately before putting the image back together.

Trapped Knowledge: Farming guidelines are stuck in heavy field manuals. **Solution:** Created an **AI Search Engine (RAG)** so workers can just type a question and instantly find the exact agricultural guideline they need.

AUTOMATED FRESHNESS GRADING AND TEA TYPE CLASSIFICATION



TeaVision

Smart Tea Estate Assistant

Home

Overview & Quick Actions

FIELD WORK

Inspect My Leaves

Scan & identify leaf types

Grade My Tea

Check freshness quality

Check for Disease

Identify leaf diseases

Identify Weeds

Spot invasive species

GUIDANCE

What Should I Do?

Treatment recommendations

Season Risk Alert

Outbreak prediction

Ask an Expert

AI chat & voice

MY ESTATE DATA

Monthly Yield Forecast

Monthly prediction

Digitize Yield Logbook

Scan handwritten records

Tea Handbook

Search tea knowledge

MARKET & AUCTION

Tea Leaf Detection

YOLOv8 + SAHI Object Detection for Tea Quality Analysis



Click or drag to upload

JPG, JPEG, PNG supported

Take Photo

From Phone

Detection Settings

DETECTION MODEL

TeaNet Roboflow v4

RECOMMENDED

Confidence Threshold: **0.35**

0.10 (Sensitive)

0.90 (Strict)

☐ Use SAHI (better for small objects)

Detect

Detection Classes

Coarse pluck

Damage Spot

Damaged Leaf

Fresh Bud 1

Fresh Bud 2

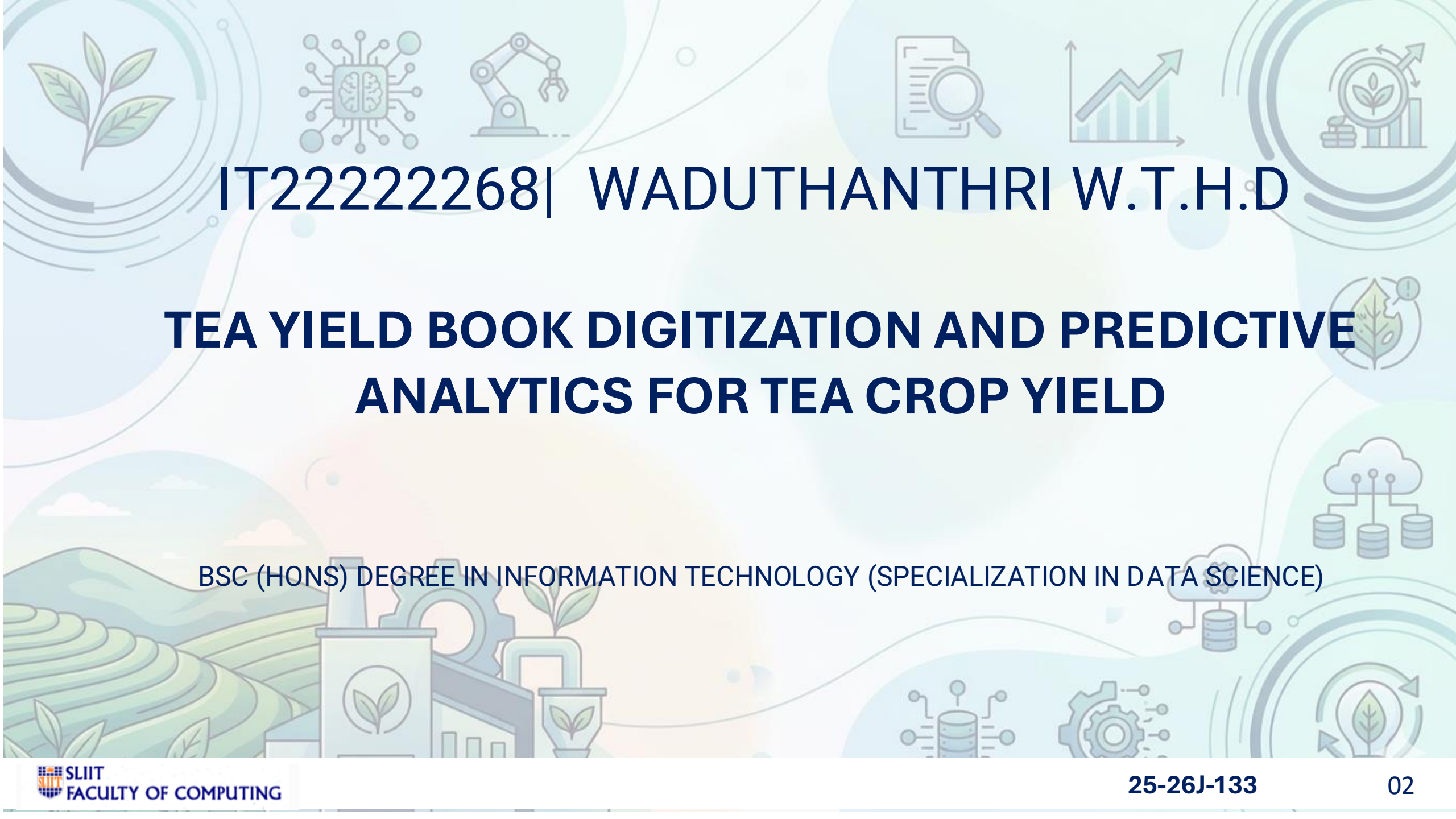
Old Leaf

stems

 **SLIIT**
FACULTY OF COMPUTING

25-26J-133

02



IT22222268| WADUTHANTHRI W.T.H.D

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY (SPECIALIZATION IN DATA SCIENCE)

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

PROBLEM: Sri Lanka's mid-country tea estates still rely on manually handwritten yield books that are vulnerable to physical damage and data loss, while simultaneously lacking any field-level yield forecasting system - leaving estates unable to preserve historical production data or make data-driven agronomic decisions.

Novelty : No prior system or research in Sri Lanka has scanned and digitized handwritten agricultural yield books from tea estates using OCR technology. also No existing yield forecasting model in Sri Lanka operates at the field level for mid-country tea estates.

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD



TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

CORE MODULES

1. OCR Model Development for Yield Book Digitization

- Using Tea Vision app capture the Tea Yield book Images and pass to Web App
- Manually labeled cells in yield book images for training
- Trained a custom PaddleOCR model fine-tuned specifically for tea estate yield book recognition

2. Monthly yield forecasting module

- Historical tea yield data collected from **2014 - 2025** (10 years)
- Data gathered from **2 mid-country estates**:
 - Hanthana Estate
 - Nayapana Estate
- Separate forecasting models trained **per divisions**

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

OCR model Accuracy

== OVERALL ==

Total cells : 739

Exact : 65.0%

Partial $\geq$ 0.5 : 91.2%

OCR Pipeline Metrics:

Threshold	TP	FP	FN	Precision	Recall	F1
0.3	709	30	0	95.9%	100.0%	97.9%
0.4	691	48	0	93.5%	100.0%	96.6%
0.5	674	65	0	91.2%	100.0%	95.4%
0.6	634	105	0	85.8%	100.0%	92.4%

Local Disk (C:) > Users > User > PaddleOCR > output > logbook_rec

Name	Date modified
best_model	3/8/2026 3:48 PM
inference	3/9/2026 12:53 AM
best_accuracy.pdopt	3/8/2026 9:34 PM
best_accuracy.pdparams	3/8/2026 9:34 PM
best_accuracy.states	3/8/2026 9:34 PM

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

Model Development

Three forecasting models were developed:

- SARIMAX
- SARIMA
- Prophet

```
{'Overall_MAE': 56.1833771650394, 'Overall_RMSE': 71.117521770148, 'Overall_MAPE': 29.398332480985392}
```

```
{'Overall_MAE': 58.84950797189577, 'Overall_RMSE': 72.0707848935334, 'Overall_MAPE': 31.711770140380203}
```

```
{'Overall_MAE': 87.9361699077162, 'Overall_RMSE': 106.88132214987382, 'Overall_MAPE': 49.16172360868863}
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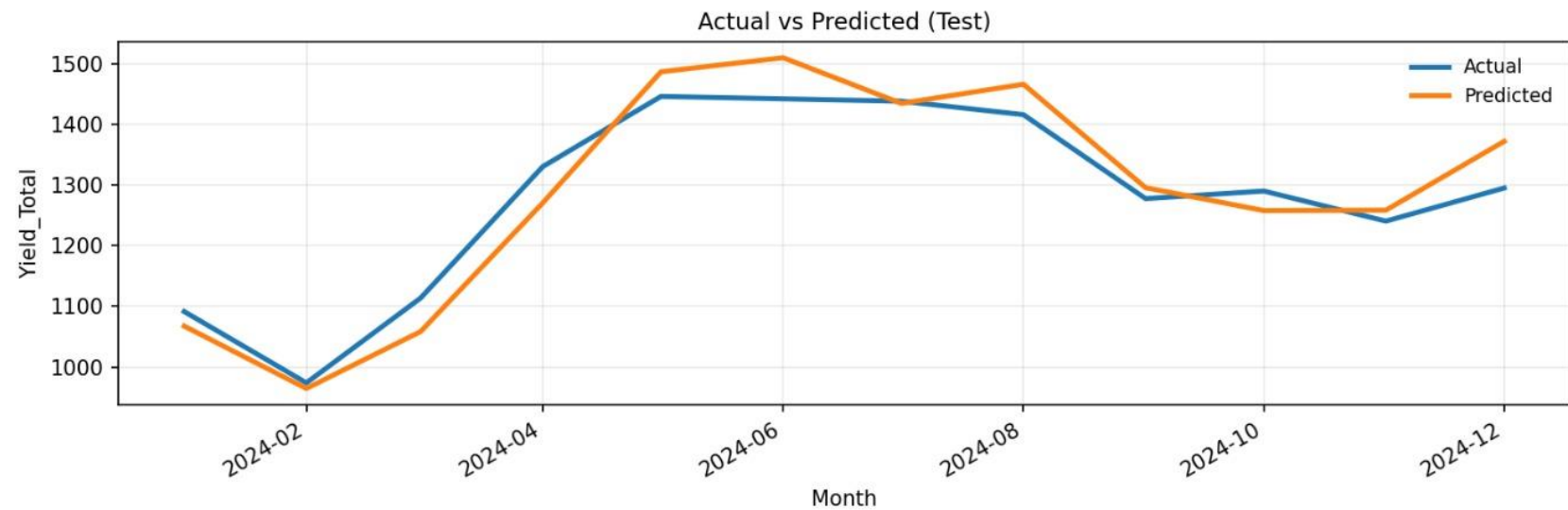
SARIMAX

SARIMA

PROPHET

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

This graph illustrates the comparison between actual observed values and SARIMAX-predicted values over time, allowing evaluation of how closely the model forecasts follow the real data trend



TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS FOR TEA CROP YIELD

Completed after PP1:

- Integrated the component with the full system
- Created the again ocr model with the cell by cell ocr
- Rebuilt and fine-tuned the models for improved accuracy
- Developed the mobile application component and integrated it into the system

Changes done for PP1 recommendations:

- Evaluated and validated OCR accuracy
- Shifted yield forecasting approach to a **time series model** as recommended by the panel

To Do

- Testing for the system
- Plan to improve speed

TEA YIELD BOOK DIGITIZATION AND PREDICTIVE ANALYTICS

The screenshot displays a web application titled "Logbook OCR Extraction" running on localhost:3000. The interface features a dark green sidebar on the left with a navigation menu. The main content area has a dark green header with the title "Logbook OCR Extraction" and a sub-header "Digitize Tea Yield Book Images". Below this, there are two tabs: "OCR Scan" (PRIMARY) and "Manual Entry" (FALLBACK). A green status bar indicates "OCR Engine ready-Upload a logbook image to extract data automatically". The central area contains a large white box with a document icon and the text "Drop logbook image here or click to upload" and "JPEG, PNG supported • Original resolution preserved for OCR". To the right of this box is a section titled "OCR Pipeline-How It Works" with a four-step process: 1. Image & Grid, 2. Cell OCR, 3. Post-Processing, and 4. Structured Output. At the bottom of the main area, there are three buttons: "QR Connect", "From Phone", and "Extract Table Data".

Navigation Menu (Left Sidebar):

- FIELD WORK
 - Inspect My Leaves (Scan & identify leaf types)
 - Grade My Tea (Check freshness quality)
 - Check for Disease (Identify leaf diseases)
 - Identify Weeds (Spot invasive species)
- GUIDANCE
 - What Should I Do? (Treatment recommendations)
 - Season Risk Alert (Outbreak prediction)
 - Ask an Expert (AI chat & voice)
- MY ESTATE DATA
 - Monthly Yield Forecast (Monthly prediction)
 - Digitize Yield Logbook** (Scan handwritten records)
 - Tea Handbook (Search tea knowledge)
- MARKET & AUCTION
 - Auction Simulation (MAS Decision System)
 - Price Forecasting (ARIMA Prediction)
- TOOLS
 - Connect Phone (Scan QR to capture)



IT22247018 | BANDARANAYAKE B.G.W.T.N

TEA MARKET DEMAND FORECASTING AND SMART AUCTION PLANNING

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY (SPECIALIZATION IN DATA SCIENCE)

Tea market demand forecasting and smart auction planning

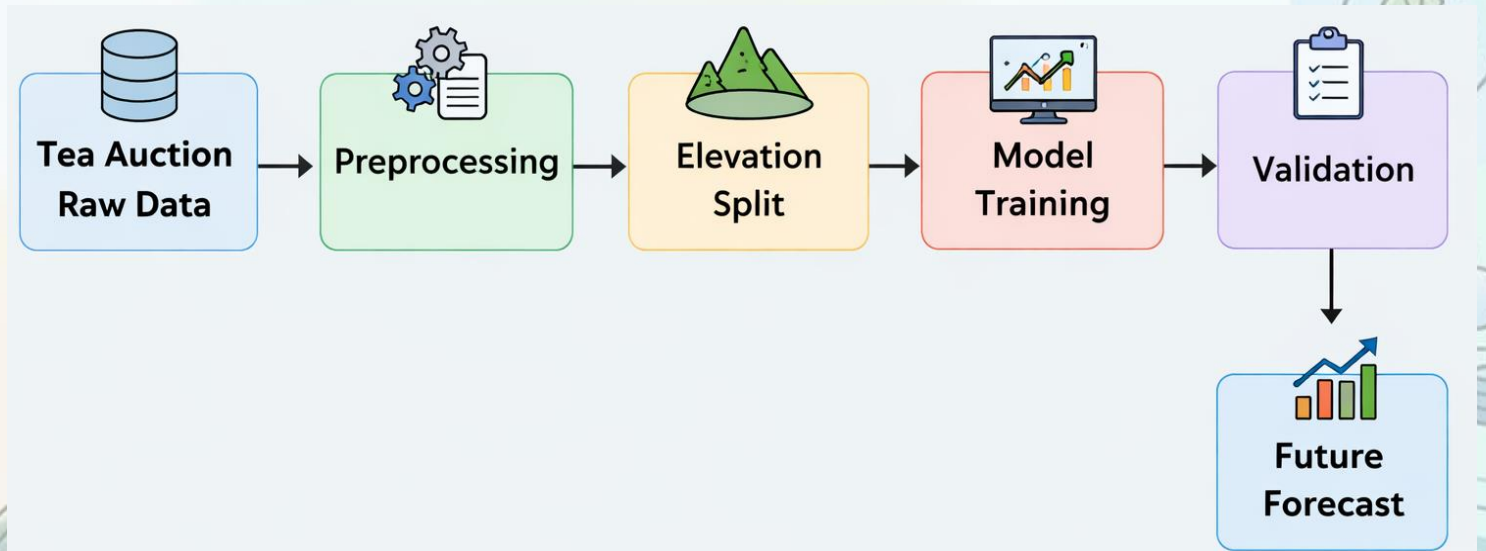
Problem: Small tea producers and factories still rely on manual estimation and guesswork for auction and production decisions. This leads to inefficient planning, supply–demand mismatches, and financial losses, while existing forecasting methods lack elevation-specific and factory-level accuracy.

Novelty : Predict weekly tea price and factory volume using localized AI models, then use multi-agent auction simulation to recommend selling, storing, and bidding decisions a capability currently unavailable in Sri Lanka's tea auction management systems.

Tea market demand forecasting and smart auction planning

Proposed Forecasting Framework

- Historical tea auction data collected from 2023–2025
- Data grouped by:
 - Low elevation
 - Mid elevation
 - High elevation
- Separate models trained per elevation



Tea market demand forecasting and smart auction planning

Model Development

Three forecasting models were developed:

- ARIMA
- SARIMAX
- Prophet

Separate models for:

- Low elevation
- Mid elevation
- High elevation

ARIMA

SARIMAX

Prophet

Tea market demand forecasting and smart auction planning

Validation Approach

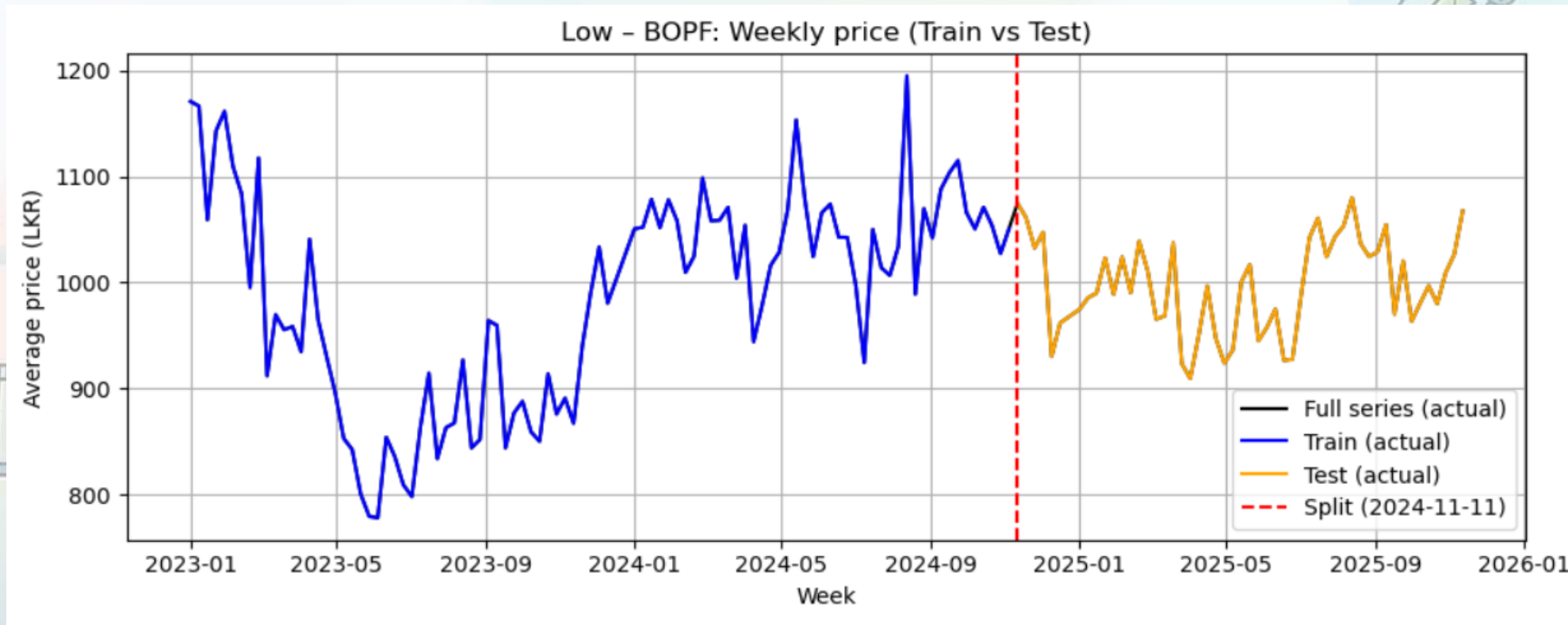
- Time-series train-test split used
- 95 weeks training
- 52 weeks testing
- Metrics:
 - MAE
 - RMSE
 - MAPE

=== Low - BOPF ===

Total points: 147

Train points: 95 (2023-01-01 → 2024-11-04)

Test points : 52 (2024-11-11 → 2025-11-12)

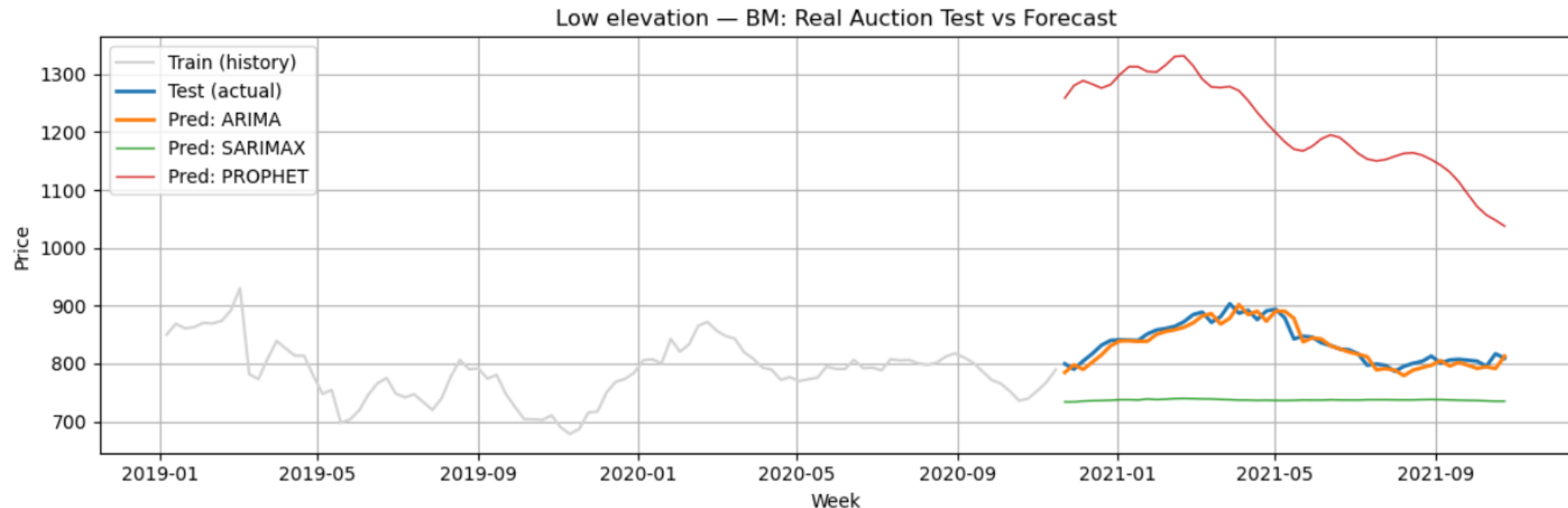


Tea market demand forecasting and smart auction planning

Validation Results Across Elevations-LOW

Elevation: Grade:
✓ BEST (lowest MAPE): ARIMA | Low - BM
Weekly points: total=147, train=98, test=49

	Model	MAE	RMSE	MAPE_%
0	ARIMA	9.834508	12.153602	1.172616
1	SARIMAX	98.928548	104.448588	11.689653
2	PROPHET	375.323402	381.327098	44.900465



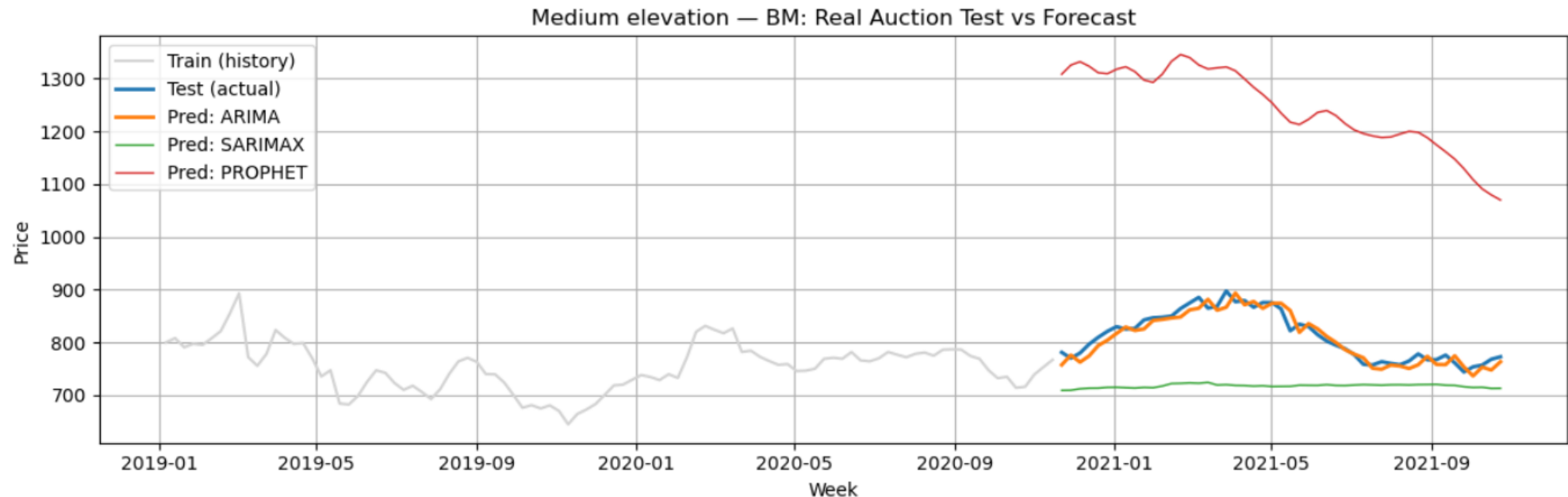
Tea market demand forecasting and smart auction planning

Validation Results Across Elevations-MID

Elevation: Medium Grade: BM

✓ BEST (lowest MAPE): ARIMA | Medium - BM
Weekly points: total=147, train=98, test=49

	Model	MAE	RMSE	MAPE_%
0	ARIMA	11.934183	14.198438	1.467254
1	SARIMAX	94.624275	104.559795	11.386664
2	PROPHET	433.026659	436.426952	53.432623

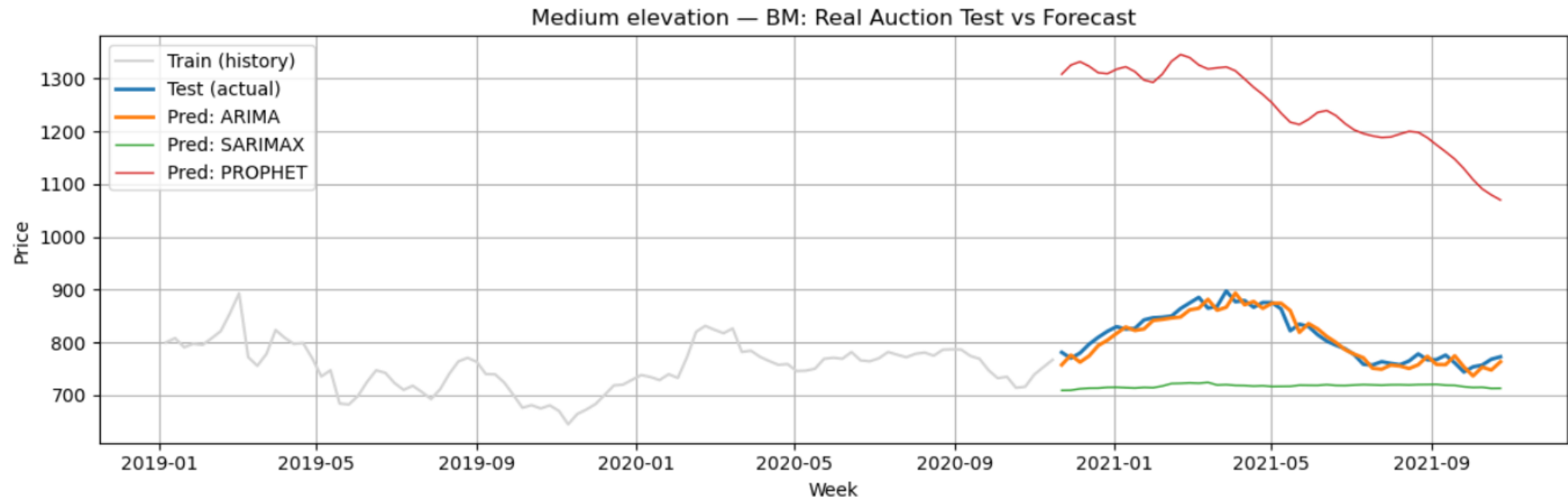


Tea market demand forecasting and smart auction planning

Validation Results Across Elevations-MID

Elevation: Medium Grade: BM
✓ BEST (lowest MAPE): ARIMA | Medium - BM
Weekly points: total=147, train=98, test=49

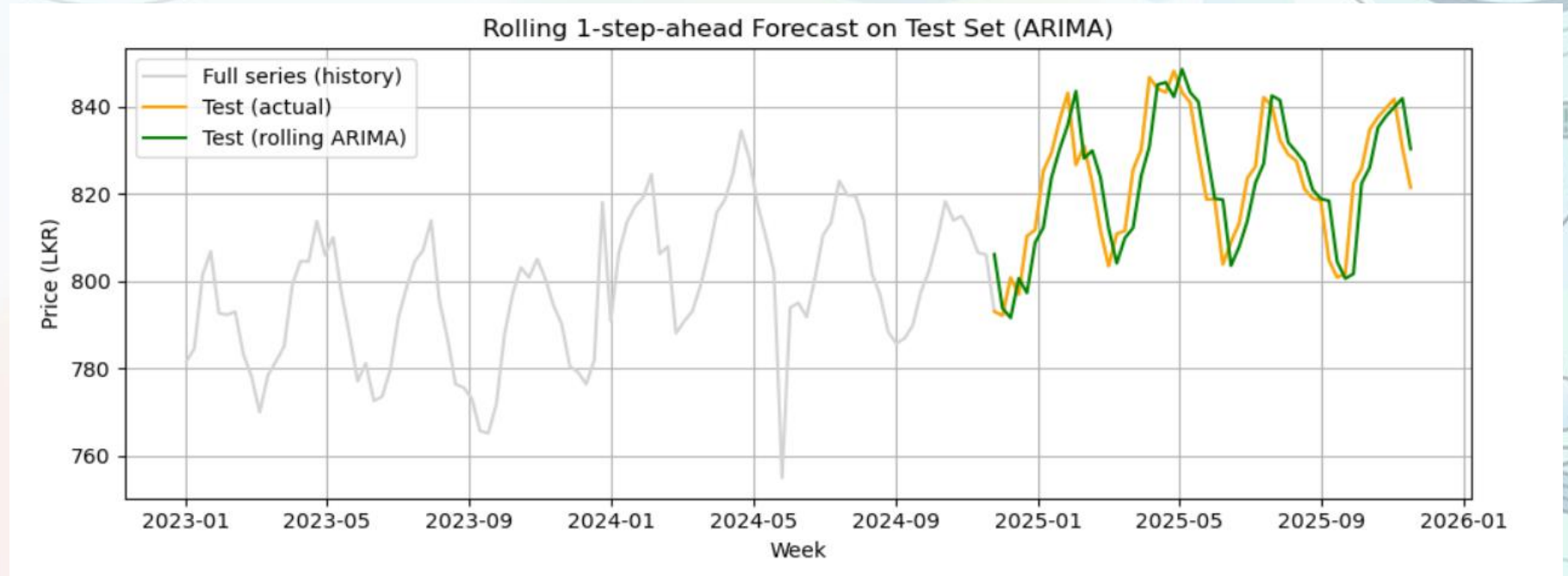
	Model	MAE	RMSE	MAPE_%
0	ARIMA	11.934183	14.198438	1.467254
1	SARIMAX	94.624275	104.559795	11.386664
2	PROPHET	433.026659	436.426952	53.432623



Tea market demand forecasting and smart auction planning

Validation Results

- **Low Elevation**
ARIMA = Best
MAPE = 1.10% – 20%
- **Medium Elevation**
ARIMA = Best
MAPE = 1.50% – 30%
- **High Elevation**
ARIMA = Best
MAPE = 1.6% – 20%



Tea market demand forecasting and smart auction planning

DEMO

Tea Broker AI System

Multi-Agent RL Auction + Tea Price Forecasting

MAS Auction SimulationTea Price Forecast

Tea Price Forecast Inputs

Elevation ⓘ

Mid

Low

Mid

High

Year ⓘ

2026

Month ⓘ

Jan

Week ⓘ

1

Show last weeks ⓘ

12 weeks

Predict Tea Price

Reload Tea Price Models

Predicted Unit Price ⓘ

Rs. 0.00

Select inputs and run prediction

Expected Price Range ⓘ

Rs. 0.00 - Rs. 0.00

No forecast yet

Tea Price Forecast Summary

Run a price prediction to see the result, target date, range and trend.

Forecast Points

Date	Week	Predicted	Low	High
No forecast points yet.				

Tea market demand forecasting and smart auction planning

Multi-Agent Tea Auction Simulation System

- MAS is a system with multiple intelligent agents
- each agent has its own role and decision
- agents interact with each other to produce a final outcome

In this project

- **Factory Agent** → decides reserve price and release quantity
- **Broker Agent** → decides market signal and commission
- **Buyer Agent** → decides bidding strength

Tea market demand forecasting and smart auction planning

Problem in Traditional Tea Auction Decision-Making

- auction decisions are often manual
- difficult to estimate correct reserve price
- hard to react quickly to changing market conditions
- no integrated system to simulate factory, broker, and buyer behavior together

Proposed MAS Solution

- input auction details
- forecast market price
- run 3 intelligent agents
- combine their decisions
- show final sale outcome and profits



Tea market demand forecasting and smart auction planning

- Factory Agent
 - sets reserve price
 - decides how much volume to release
- Broker Agent
 - gives market signal
 - selects commission guidance
- Buyer Agent
 - decides how strongly to bid

- **ARIMA** for tea price forecasting
- **DQN**(Deep Q-Network) for Buyer Agent
- **PPO**(Proximal Policy Optimization) for Factory Agent
- **PPO**(Proximal Policy Optimization) for Broker Agent
- **FastAPI** for backend API
- **React** for frontend dashboard

Tea market demand forecasting and smart auction planning

Agent Roles

•Factory Agent (PPO)

- Low Reserve** → easier sale, lower minimum price
- Balanced Reserve** → balance between profit and sale chance
- High Reserve** → higher minimum price, stronger profit target
- Release Factor** → how much of the lot is released now

•Buyer Agent (DQN)

- Wait / Hold** → not bidding strongly now
- Cautious Bid** → careful bidding
- Competitive Bid** → stronger bid to improve winning chance
- Very Aggressive Bid** → strongest bid to secure the lot quickly

•Broker Agent (PPO)

- Bullish** → market likely rising
- Neutral** → market stable or unclear
- Bearish** → market likely falling
- Commission Rate** → broker fee from successful sale

Tea market demand forecasting and smart auction planning

Validation Results

- Buyer directional accuracy = **100.0%**
- Broker directional accuracy = **99.9%**
- Factory reserve decision accuracy = **81.7%**
- Factory profitable MAS rounds = **73.0%**
- Broker profitable MAS rounds = **73.0%**

How the system was validated

- Validation was done using real historical tea auction transition data
- Total real transition rounds used: **2,743**
- Directional evaluation rounds: **2,229**
- Full MAS evaluation rounds: **2,229**

```
TRAIN_STEPS = 600_000
best_dir = "./models/buyer_best/"
os.makedirs(best_dir, exist_ok=True)
```

This accuracy validation was performed using:

1. Real historical transition rounds : 2,743
2. Directional evaluation rounds : 2,229
3. Full MAS evaluation rounds : 2,229

Key measured results:

- Buyer directional accuracy : 100.0%
- Broker directional accuracy : 99.9%
- Factory reserve decision accuracy : 81.7%
- Factory profitable MAS rounds : 73.0%
- Broker profitable MAS rounds : 73.0%

Overall research/demo score : 85.5%

Overall grade : STRONG

Final verdict:

Strong research/demo validation performance.

Tea market demand forecasting and smart auction planning

DEMO

Tea Broker AI System

Multi-Agent RL Auction + Tea Price Forecasting

MAS Auction SimulationTea Price Forecast

Inputs

Lot Volume ⓘ

Factory Elevation ⓘ

Tea Grade ⓘ

☐ Use Current Market Price ⓘ

Market Demand ⓘ

Competition ⓘ

Storage Cost (Rs/kg/week) ⓘ

☐ Set Production Cost ⓘ

Period ⓘ
Year ⓘ
Month ⓘ
Week ⓘ

TEA MARKET DEMAND FORECASTING AND SMART AUCTION PLANNING

Completed after PP1:

1. Tea Auction Forecasting Model

Developed weekly tea price forecasting models using large-scale Colombo Tea Auction data.

Trained using **around 600,000+ auction records** with grade, elevation, quantity, buyer, and broker information.

Supports **Low, Mid, and High elevation prediction**.

2. Smart Auction Behavior Simulation

Designed a **Multi-Agent System** to simulate real tea auction interactions.

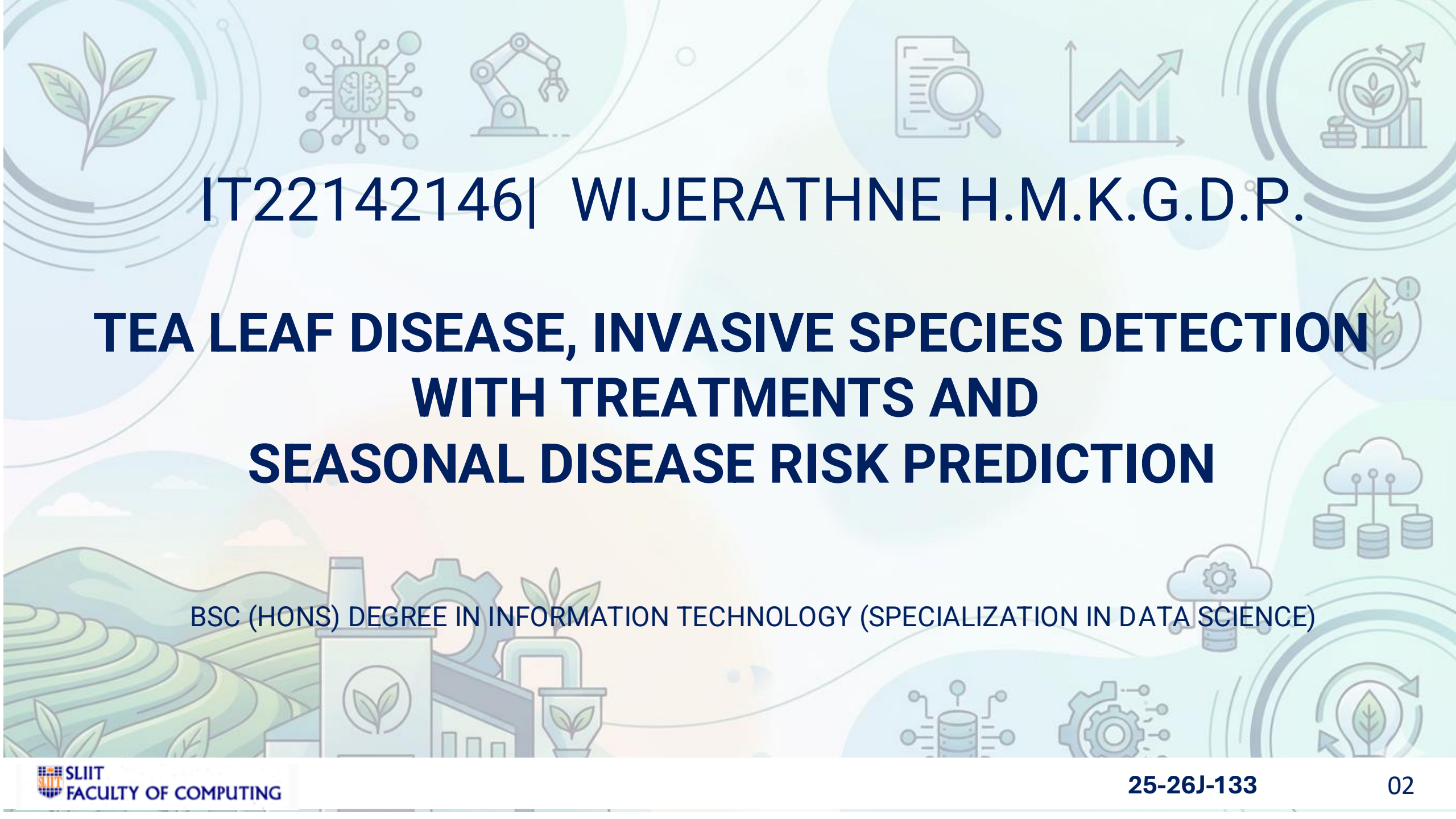
Includes **Factory Agent, Broker Agent, and Buyer Agent** for selling, bidding, and market strategy decisions.

Provides realistic auction outcome prediction and scenario testing.

3. Full System Development and Integration

Integrated forecasting and auction simulation into a complete web-based platform.

Developed **React frontend interfaces**, backend APIs, dashboards, and result visualization components for end users.



IT22142146| WIJERATHNE H.M.K.G.D.P.

TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY (SPECIALIZATION IN DATA SCIENCE)

TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

PROBLEM:

Tea farmers and estate managers struggle to protect crop yields due to a slow, manual identification of leaf diseases and invasive plants. Without immediate, accurate and tailored treatment guidance, diseases spread rapidly, reducing tea quality. Furthermore, farmers lack early-warning systems to prepare for climate-driven seasonal outbreaks, forcing them to react to damage rather than prevent it.

Novelty :

- **Explainable Visual AI:** Uses CNNs combined with Grad-CAM to visually highlight the exact disease or pest location on the leaf, building farmer trust.
- **Climate-Linked Forecasting:** Predicts seasonal disease risks (Low/Medium/High) by analyzing environmental data like temperature and rainfall.
- **Intelligent Advisory:** Instantly provides expert-verified treatment plans (organic/chemical) and features a voice-enabled NLP chatbot for real-time assistance.



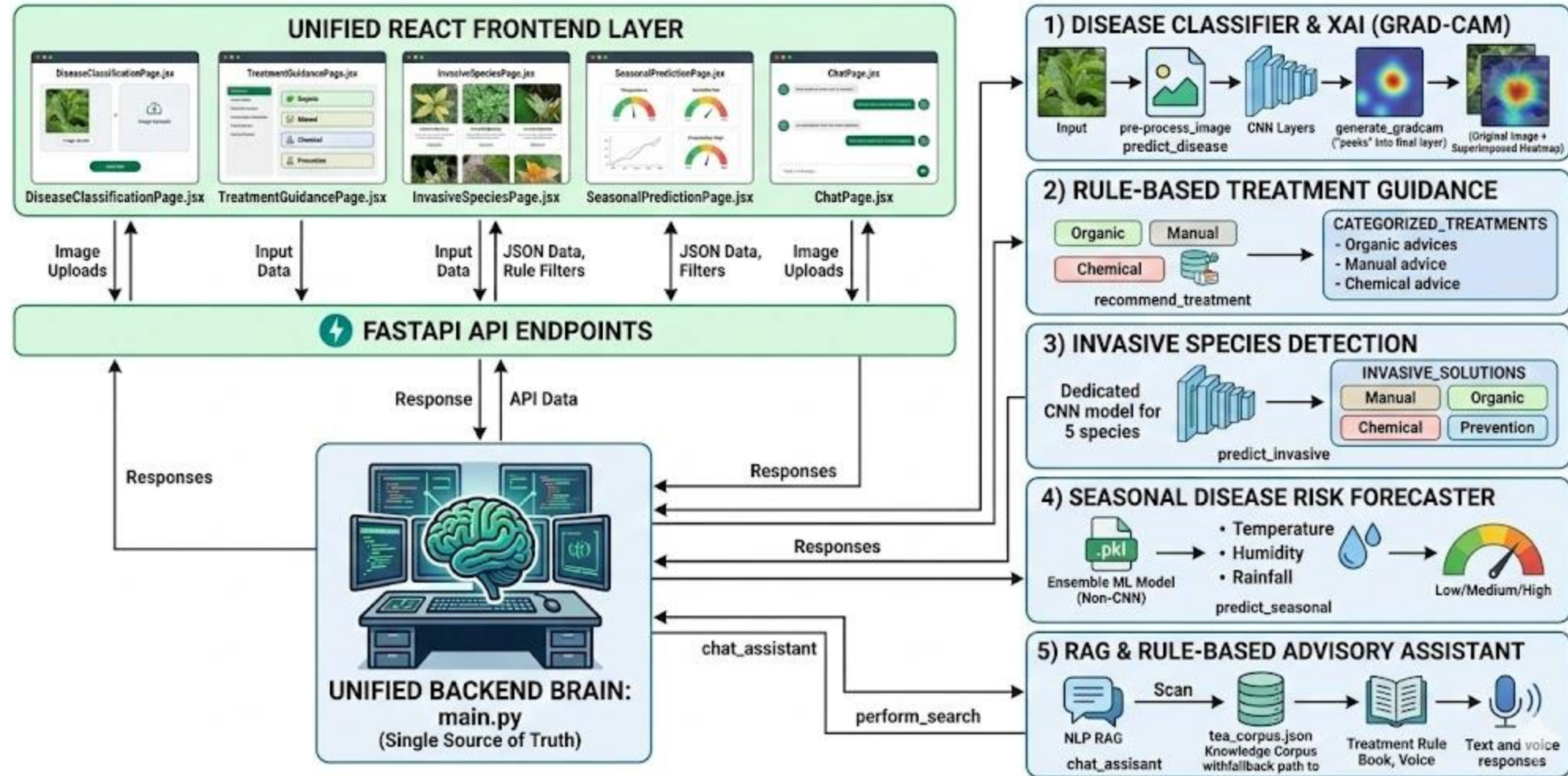
TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

Core Modules:

- Tea Leaf Disease Classification (CNN + XAI Grad-Cam)
- Invasive Species Identification
- Knowledge based Treatment Recommendation System
- Climate-Linked Seasonal Disease Prediction
- NLP & Speech AI Farmer Assistant

TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND

T-AI: UNIFIED TEA HEALTH DIAGNOSTIC ECOSYSTEM - ADVANCED SYSTEM ARCHITECTURE

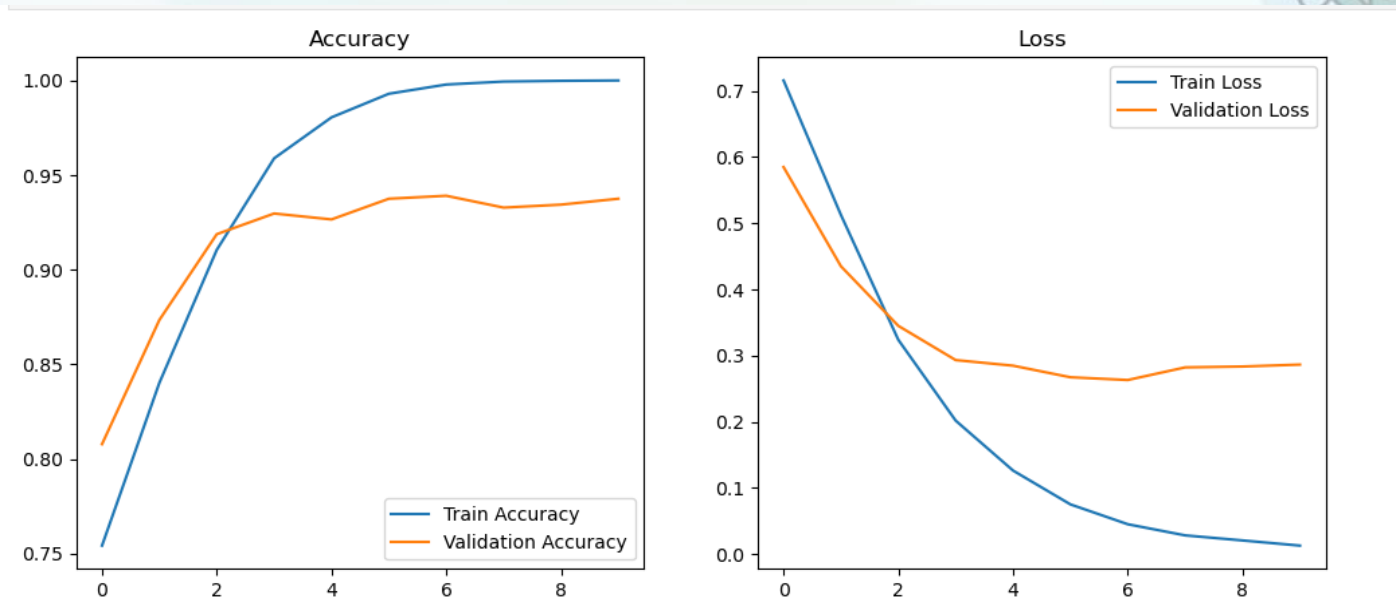


TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

Tea Leaf & Pest Disease Detection Model (CNN)

- Convolutional Neural Network for image classification of tea leaf diseases.
- Dataset Size: ~6,000 tea leaf images.
- Diseases Detected: Healthy, Brown Blight, Grey Blight, Red Rust, Blister Blight, Helopeltis
- **Accuracy: ~92%**

21/21 ————— 18s 197ms/step - accuracy: 0.9390 - loss: 0.1743
Test Accuracy: 0.9389880895614624

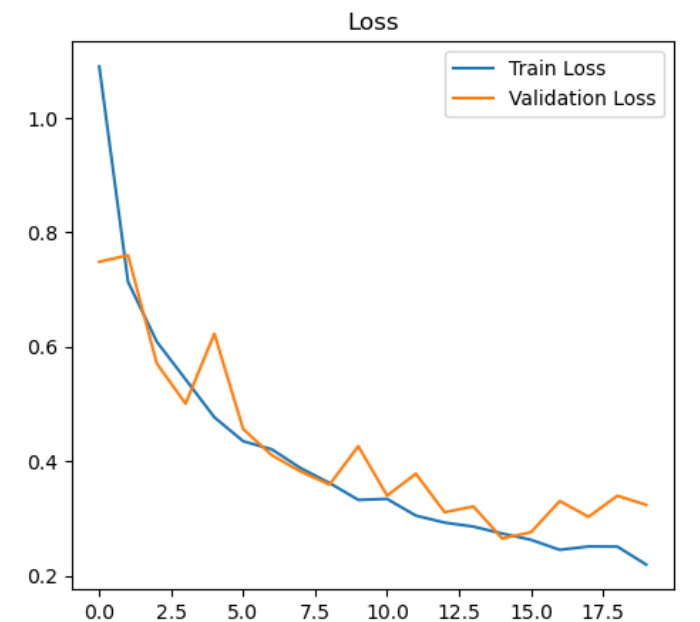
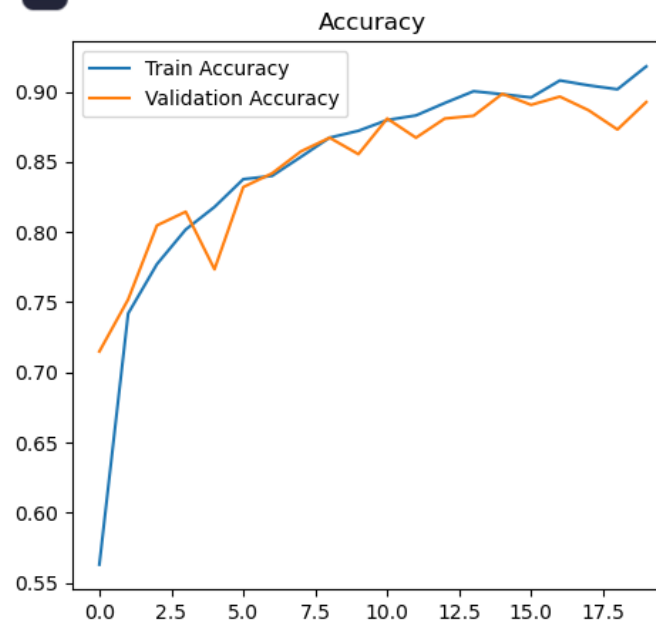


TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

Invasive Species Detection Model (CNN)

- CNN-based model for identifying invasive species on tea plant.
- Dataset Size: ~5,000 pest-affected tea leaf images.
- Invasive Species Detected: Lantana camara, Tridax procumbens, Mikania micrantha, Mimosa diplotricha, Sphagneticola trilobata).
- **Accuracy: ~91%**

18/18 — 18s 257ms/step - accuracy: 0.9149 - loss: 0.2500
Test Accuracy: 0.9149305820465088



TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

- Ensemble machine learning model combining Random Forest and LightGBM was developed to predict seasonal tea diseases based on environmental factors such as month, season, region, temperature, humidity, and rainfall.
- Dataset Size: ~15,300 environmental records.
- Diseases Predicted: Healthy, Brown Blight, Grey Blight, Red Rust, Helopeltis, Blister Blight.
- **Accuracy: ~85%**

Classification Report:

	precision	recall	f1-score	support
Blister Blight	0.82	0.94	0.88	444
Brown Blight	0.67	0.66	0.66	618
Grey Blight	0.69	0.61	0.65	496
Healthy	0.73	0.54	0.62	599
Helopeltis	0.84	0.95	0.89	412
Red Rust	0.68	0.82	0.74	491
accuracy			0.73	3060
macro avg	0.74	0.75	0.74	3060
weighted avg	0.73	0.73	0.73	3060

TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

NLP Tea AI Assistant (Tea Knowledge Chatbot)

- NLP-based chatbot developed to provide interactive guidance on tea cultivation, diseases, pests, and treatments.
- Knowledge Source: Uses a tea knowledge database containing curated information on tea diseases, pest management, and treatment practices.
- Core Processing: Implements a keyword-based search mechanism where the backend analyzes the user's query and retrieves the most relevant answer from the knowledge corpus.
- Voice Interaction: Supports voice interaction (speech-to-text and text-to-speech) and integrates Web Speech API, using SpeechRecognition for converting speech to text and SpeechSynthesis for reading AI responses aloud.
- Additional Features: Supports quick navigation links to related modules such as disease detection and treatment guidance, enabling seamless interaction within the system.

TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

Completed after PP1:

1. Invasive Species Detection Model

Developed a second CNN trained on 5,000+ images to detect 5 common invasive weeds. (Lantana camara, Tridax procumbens, Mikania micrantha, Mimosa diplotricha, and Sphagneticola trilobata)

Accuracy: 91.49%

2. Seasonal Disease Prediction

Analyzed 15,000 environmental records (Temperature, Humidity, Rainfall, Season, Region).

Deployed Random Forest & Gradient Boosting models for risk-level predicting.

Accuracy: 73%

3. NLP Farmer Assistant

Integrated a voice-and-text AI chatbot for real-time, interactive treatment guidance.

Changes done for PP1 recommendations:

- Improved Model Evaluation: Performed detailed evaluation of all AI models and optimized them to achieve higher prediction accuracy and reliability.
- Added additional intelligent components such as seasonal disease prediction, invasive species detection, and an NLP-based assistant etc.

To Do:

Testing for the system

TEA LEAF DISEASE, INVASIVE SPECIES DETECTION WITH TREATMENTS AND SEASONAL DISEASE RISK PREDICTION

The screenshot displays the 'Tea Analysis System' dashboard. The browser's address bar shows 'localhost:3000'. The dashboard has a dark green sidebar on the left with a 'Tea Analysis' header and a list of components: Dashboard, Leaf Detection, Quality Grading, Knowledge Base, Disease Classify, Invasive Species, Seasonal Predict, Treatment Guide, and Tea AI Assistant. The main content area features a header 'AI-Driven Tea Production & Quality System' and four large white boxes with metrics: 10 AI MODELS, 7 DETECTION CLASSES, 156 KNOWLEDGE DOCS, and 8 COMPONENTS. Below these are four detailed component cards: 'Leaf Detection' (YOLOv8 + SAHI), 'Automated Freshness Grading & Classification' (10 classical ML models), 'Tea Knowledge RAG System' (Retrieval-Augmented Generation), and 'Seasonal Disease Prediction' (Predictive analytics). Each card includes a description, a list of technologies used, and an 'Open' button.

Tea Analysis
Project 25-26J-133

AI-Driven Tea Production & Quality System
Project 25-26J-133 | BSc (Hons) IT - Data Science | SLIIT

10
AI MODELS

7
DETECTION CLASSES

156
KNOWLEDGE DOCS

8
COMPONENTS

Dashboard
System Overview

COMPONENTS

- Leaf Detection**
YOLOv8 + SAHI
- Quality Grading**
ML Classification
- Knowledge Base**
RAG Search
- Disease Classify**
Identify Diseases
- Invasive Species**
Detect Species
- Seasonal Predict**
Predict Diseases
- Treatment Guide**
Disease Treatments
- Tea AI Assistant**
Ask NLP Chatbot

Research Components

Leaf Detection
YOLOv8 deep learning model with SAHI for detecting 7 tea leaf classes including damage spots, diseased leaves, and fresh buds.
YOLOv8 SAHI OpenCV FastAPI
[Open →](#)

Automated Freshness Grading & Classification
10 classical ML models extracting 25 hand-crafted features (color, texture, shape) to classify tea leaf quality as high or medium.
Scikit-learn MLP SVM AdaBoost
[Open →](#)

Tea Knowledge RAG System
Retrieval-Augmented Generation system with 156 curated documents, BM25 + Sentence-BERT hybrid search across 13 categories.
FAISS BM25 Sentence-BERT React
[Open →](#)

Seasonal Disease Prediction
Predictive analytics for tea seasonal disease risk based on seasonal data.
Disease Risk Seasonal Analytics
[Open →](#)



THANK YOU

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